1

func. $\forall x \in D,\ \underbrace{P(x)}_{①}$ → 一个 statement ②

E.g. (1) $\forall x \in \mathbb{R},\ P(x) \Leftrightarrow \sin x = 0$

~~?~~ ~~②~~

if $x = 0$: $\sin 0 = 0 \Rightarrow P(0)$ is true

if $x = \frac{\pi}{4}$: $\sin\frac{\pi}{4} = \frac{\sqrt{2}}{2} \neq 0 \Rightarrow P(\frac{\pi}{4})$ is false

(2) $\forall x \in D,\ \overset{(\neg Q(x))}{P(x)}$ ($\Leftrightarrow$ is true)

↓

(neg.) $\exists x \in D$, s.t. $\underset{(\neg Q(x))}{\neg P(x)}$ ($\Leftrightarrow$ is false)

(3) 证: $\exists x \in D$, s.t. $Q(x)$

↪ show: 取一 $x_0 \in D$, $Q(x_0)$ → (true) example ②

(4) $\forall x \in P\ P(x)$

↓

(counterex.) $\exists x_0 \in D,\ \neg P(x)$

(5) $\forall n \in \mathbb{Z}$, n is odd $\Rightarrow n^2$ is odd

↓

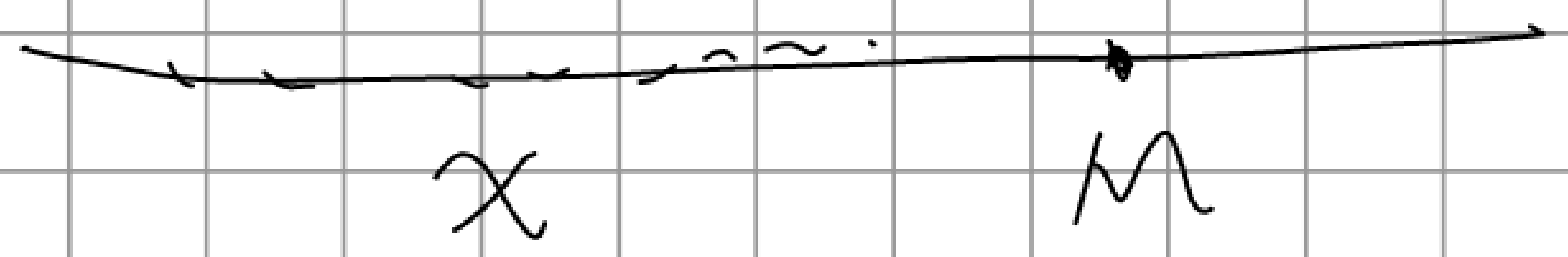
(neg.) $\exists n \in \mathbb{Z}$, s.t. n is odd $\Rightarrow n^2$ is even proper order

$\forall \alpha$ ..., $\exists \beta$, s.t. $\forall \gamma$, $\exists \delta$, s.t. ...
(?)

$(\beta = \beta(\alpha),\ \delta = \delta(\alpha, \beta, \gamma))$

E.g. let set $S \subset \mathbb{R}$

(1) $\exists M \in \mathbb{R}$, s.t. $\forall x \in S$, $x \leq M$
(M is an upper bd. of S)



(2) $\forall x \in S$, $\exists M(x) \in \mathbb{R}$, s.t. $x \leq M(x)$

let $x \in S$,

Take $M(x) = x$, $x \leq x$ ✓
(?)

(1) <=> (3) $\forall x \in S$, $\exists M \in \mathbb{R}$,
s.t. $x \leq M$ and M does not dep. on x

(?)

E.g.: There is a prime p betw. every pos. int. n and $2n$

proper order: $\forall n \in \mathbb{N}^*, \exists \text{prime } p$, s.t. $\underbrace{n \le p \le 2n}_{\text{(and)}}$

neg.: $\exists n \in \mathbb{N}^*$, s.t. $\forall \text{prime } p$, $p < n$ or $p > 2n$

E.g.: $\forall x \in \mathbb{R}$ for which $\exists y \in \mathbb{R}$, s.t. $x = y^2$, $x \geq 0$

$\Longleftrightarrow \forall x \in \mathbb{R}, \exists y \in \mathbb{R}$, s.t. $x = y^2 \Rightarrow x \geq 0$

neg.: $\exists x \in \mathbb{R}$, s.t. $\exists y \in \mathbb{R}$, s.t. $x = y^2$ and $x < 0$

neg.: $\exists x \in \mathbb{R}$ for which $\exists y \in \mathbb{R}$, s.t. $x = y^2$, $x < 0$

$f(x)$ is cont. at $a \in B$ $\nearrow$ on D

if $\forall \varepsilon > 0, \exists \delta > 0$,

s.t. $\underbrace{\ }_{\forall x \in D,}$ if $|x - a| < \delta$, then $|f(x) - f(a)| < \varepsilon$

neg. $\exists \varepsilon > 0$, s.t. $\forall \delta > 0$, $\exists x \in \mathbb{R}$,

s.t. $|x - a| < \delta$ and $|f(x) - f(a)| \geq \varepsilon$

$f(x) \to 10$
$x \to \infty$

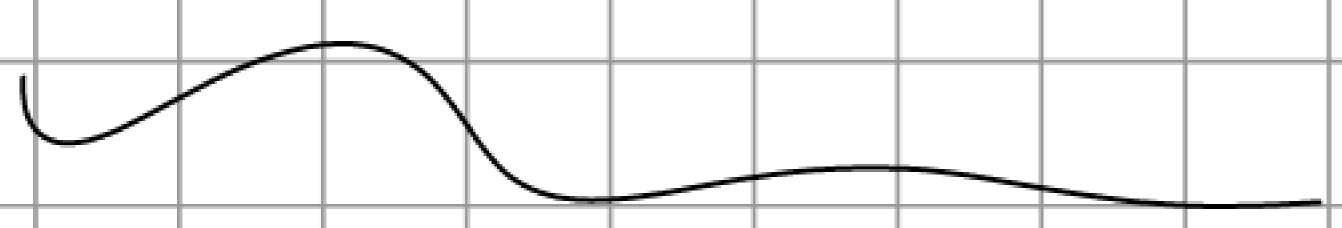
$\forall \varepsilon > 0, \exists M \in \mathbb{R}$,

s.t. $\forall x \in \mathbb{R}$, if $x \geq M$ then $|f(x) - 10| < \varepsilon$

method of exhaustion

② S limit. S $x_1, \cdots, x_n$

—— (见图)



3

$x^5 + 8x^3 + 1 = 0$ has no real root

$\Leftrightarrow \forall x \in \mathbb{R},\ x^5 + 8x^3 + 1 = 0 \Rightarrow x < 0$

$\Leftrightarrow$ (contrapos.) $\forall x \in \mathbb{R},\ x \geq 0 \Rightarrow x^5 + 8x^3 + 1 \neq 0$

Let $x \in \mathbb{R}$

Assume $x \geq 0$

Then $x^5 \geq 0$ and $8x^3 \geq 0$

so $\underbrace{x^5 + 8x^3 + 1}_{\downarrow} > 0$

$\Rightarrow [\ ] \neq 0$

By contradiction

Assume the contrary $\perp$(?)

Then $\exists x \in \mathbb{R}$, s.t. $x^5 + 8x^3 + 1 = 0$ and $x \geq 0$

$x \geq 0 \Rightarrow x^5 + \sim > 0 \Rightarrow x^5 + \sim \neq 0$, a contradiction

(② str.)

t^2 is ↑ (increasing) for $t \geq 0$

Lemma: Let $x, y \in \mathbb{R}$, s.t. $x, y \geq 0$

Then $x \leq y \iff x^2 \leq y^2$

Proof: Assme $x, y \geq 0$

(证 ⇒:)

Assme $x \leq y$ (条件)

① $x^2 \leq xy$

② $xy \leq y^2$

By transitivity, $x^2 \leq y^2$

(证 ⇐:)

Assme $(0 \leq)\, y < x$, ③ In particular $x > 0$

① by $x > 0$, $xy < x^2$

② by $y \geq 0$, $y^2 \leq xy$

Thus, $y^2 < x^2$, contradiction

(反证)

~ ~ ~ (終了)

AM - GM inequality

$\forall x, y \in \mathbb{R}$ and $x, y \geq 0$, $\sqrt{xy} \leq \frac{x+y}{2}$

$\sqrt{xy} = \frac{x+y}{2} \iff x = y$

Proof:

Let $x, y \geq 0$

$$0 \leq \sqrt{xy} \leq \frac{x+y}{2} \overset{\text{(Lemma)}}{\iff} xy \leq \left(\frac{x+y}{2}\right)^2$$

$$\iff 4xy \leq (x+y)^2$$

$$\iff 4xy \text{ -- -}$$

$$\iff 0 \leq x^2 - 2xy + y^2$$

$$\iff \underbrace{(x-y)}_{\text{①} \in \mathbb{R}}^2 \text{, True}$$

(証 $\sim = \sim \iff \sim = \sim$, 同様: $\sim \iff (x-y)^2 = 0 \iff x = y$

tri. inequality

$\forall x, y \in \mathbb{R}$

$|x+y| \leq |x| + |y|$

Proof:

$|x+y| \leq |x| + |y|$

(lemma) $\rightarrow \Longleftrightarrow$

- —— ([illegible])

Lemma: $\forall n \in \mathbb{Z}$, n^2 is even $\Rightarrow$ n is even

Theorem: $\sqrt{2}$ is irrational

Proof:

Assume $\sqrt{2}$ is rational

So $\exists m, n \in \mathbb{N}$, s.t. $\sqrt{2} = \frac{m}{n}$

W.L.O.G, we may assume that

1 is the greatest common factor

of m and n

(otherwise, — - - __②__)

Then $2 = \frac{m^2}{n^2}$, so $m^2 = 2n^2$

$\Rightarrow m^2$ is even $\overset{\text{(lemma)}}{\Longrightarrow}$ m is even

By def., ⌣— ([illegible])

$(\exists j, m = 2j \Rightarrow 4j^2 = 2n^2 \Rightarrow n^2 = 2j^2$ is even

$\Rightarrow$ n is even, __☺__ contradicts)

A prime is an int. __②__, n > 1, s.t. __②__ the only positive factors are 1 and n

Theorem: There are infinite many primes

Proof:

Assume ~ only finite many:

$\exists n$, s.t. $p_1, \cdots, p_n$ (~~①(还是有 prime?~~ ②))

Claim: $\forall k \in \mathbb{Z}$ and $k > 1$,

k has at least 1 prime factor

If k is not a prime:

Let $j = \min \{ d \geq$ ② $: d \mid k \}$

Then j is prine and $j \geq 2$

([illegible])